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Experiment 2

Measurement of g

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Worksheet

I. DERIVATION

a. Acceleration due to gravity of a system in free fall is equivalent to the difference of average velocities of the system within two time intervals divided by the average time spent by the system in free fall within these two time intervals.

$$a_g = \frac{v_{avg}^{T_2} - v_{avg}^{T_1}}{t_{avg}}$$

$$a_g = \frac{v_D^{T_2} - v_d^{T_1}}{\frac{T_1 + T_2}{2}}$$

$$g = \frac{2}{T_1 + T_2}\left(\frac{D}{T_2} - \frac{d}{T_1}\right)$$

II. PLOTS

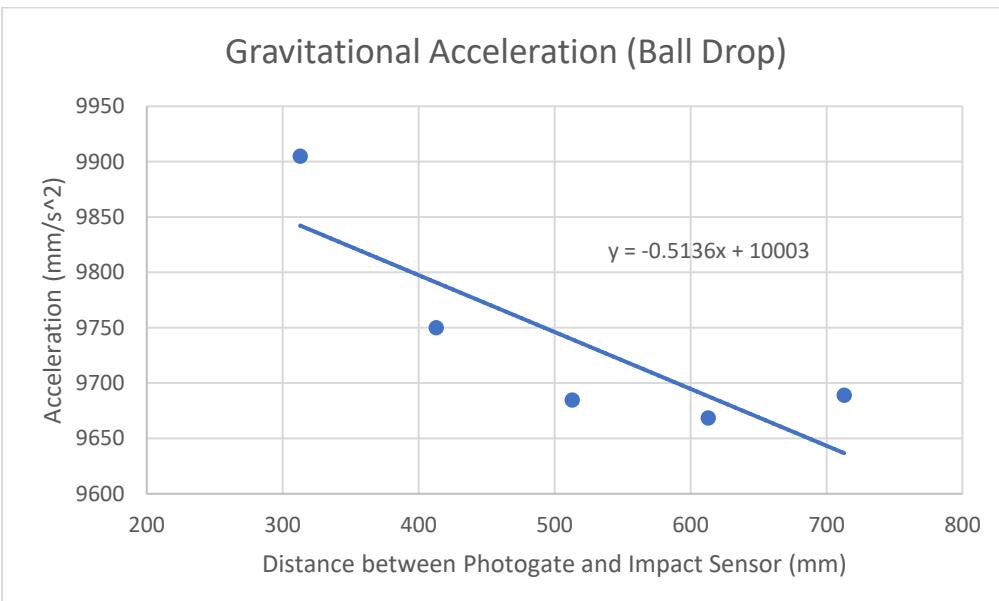

Figure 1.1: graph of measured acceleration values of a system in free fall from different measured heights (d = 313 mm, 413 mm, 513 mm, 613 mm, 713 mm). The best-fit line yields a value of $g = -0.5136D + 10003$.

The equation of the best-fit line of the graph indicates that acceleration due to gravity decreases at a factor of a half as the distance between the lower photogate and the impact sensor increases. This result conflicts with the expectation that acceleration due to gravity remains constant regardless of the height of release of a free-falling object. Therefore, the data quantitatively confirms that gravity is slightly dependent on the height of release of a falling object due to the negative slope of the best-fit line.

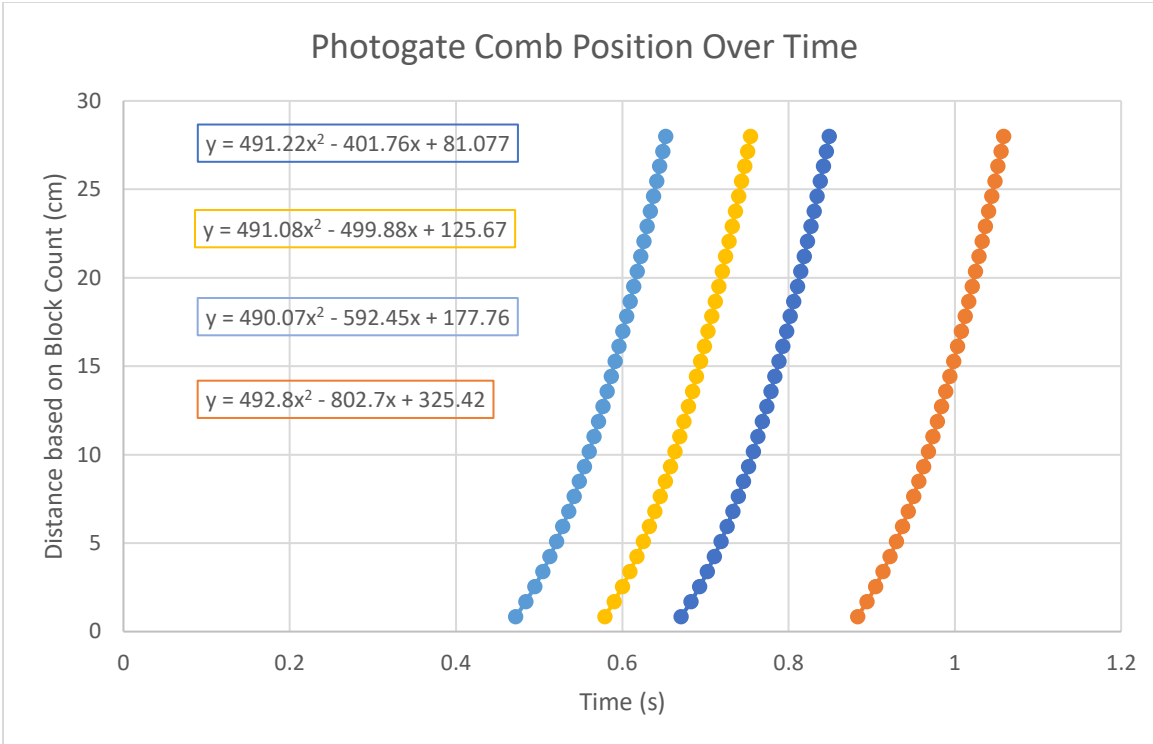

Figure 1.2: Distance the photogate comb covers over time as it passes through the photogate. All four runs of the photogate comb experiment yield similar best-fit polynomial lines of order 2. The best-fit lines yielded g_{best}, which is represented by twice the coefficient of the second-degree variable in the trendline equations because the second order polynomial is equivalent to $y = ax^2 + bx + c$ is equivalent to $x = \frac{1}{2}at^2 + vt + x_0$. Hence, $g = 2a$ and $\delta g = 2\delta a$.

Second order constant a (cm/s^2)	g (m/s^2)
490.0714 ± 0.0982	9.8014 ± 0.0020
492.8016 ± 0.5689	9.8560 ± 0.0114
491.0756 ± 0.7212	9.8215 ± 0.0144
491.2247 ± 0.1089	9.8245 ± 0.0022

Table 1.1: Data table of calculated g_{best} and δg using quadratic regression analysis. The calculations yield a value of $g = (9.8259 \pm 0.0075)\ m/s^2$.

III. DATA TABLE

Trial	Photogate spacing d (cm)	Gap to impact sensor D (cm)	Measured acceleration g (m/s^2)
1	8.7 ± 0.05	31.3 ± 0.05	9.9049 ± 0.2926
2	8.7 ± 0.05	41.3 ± 0.05	9.7502 ± 0.0288
3	8.7 ± 0.05	51.3 ± 0.05	9.6845 ± 0.0648
4	8.7 ± 0.05	61.3 ± 0.05	9.6684 ± 0.1145
5	8.7 ± 0.05	71.3 ± 0.05	9.6890 ± 0.1724

Table 1.2: Measured values of the distance regions between the photogates and the impact sensor with uncertainty from the ruler. The acceleration of the object due to gravity was calculated $g = (9.6890 \pm 0.1346)\ m/s^2$, for each heights of release D, using the equation:

$$g_{best} = \frac{1}{N}\sum_{i=1}^{N} g_i$$

Uncertainty analysis of acceleration of the measured acceleration g was calculated using the sum of the systematic and statistical uncertainties:

$$\delta g_{systematic} = \frac{g_{max} - g_{min}}{2}$$

$$\delta g_{statistical} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(g_i - g)^2}$$

$$\delta g_{total} = \frac{g_{max} - g_{min}}{2} + \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(g_i - g)^2}$$

Gap to impact sensor D (cm)	Systematic Uncertainty (m/s^2)	Statistical Uncertainty (m/s^2)	Total (m/s^2)
31.3 ± 0.05	0.1840	0.1085	0.2926
41.3 ± 0.05	0.0168	0.0121	0.0288
51.3 ± 0.05	0.0390	0.0257	0.0648
61.3 ± 0.05	0.0679	0.0466	0.1145
71.3 ± 0.05	0.1075	0.0649	0.1724

Table 1.3: Uncertainty Analysis of the acceleration due to gravity of a free-falling system. By comparing the values of the systematic and statistical uncertainties, the systematic uncertainties dominate the total uncertainties in all cases of D.

IV. CONCLUSION

The ball drop experiment yielded a final result of $g = (9.6890 \pm 0.1346)\ m/s^2$. This result falls within the expected value of gravity in Knudsen 1-238 of $g = (9.7955 \pm 0.0003) m/s^2$. The uncertainty analysis for the ball drop experiment analyzes and sums the values of the systematic and statistical uncertainties of the experiment. Table 1.2 shows that, for all measured values of distance between the photogate gap and the impact sensor D, the systematic uncertainties of each trial dominate the total uncertainty of each trial. However, the difference between the systematic and statistical uncertainties are not significant enough to treat the statistical uncertainty values as negligible; therefore, I chose a combination of both systematic and statistical error for δg. The dominance of the systematic uncertainties merely imply that the ball drop experiment data was more precise than accurate.

The photogate comb experiment yields a final result of $g = (9.8259 \pm 0.0075)\ m/s^2$, which is closer to the true value of g than the result from the ball drop experiment. Similar to the result of the ball drop experiment, the result falls within the expected value of g. However, unlike the ball drop experiment, the uncertainty analysis of the photogate comb experiment data uses the data regression tool in Excel to perform a quadratic regression. This method of uncertainty analysis calculates a near-zero margin of error. This near-zero uncertainty supports the conclusion that the photogate comb is the better method for calculating g in terms of both accuracy and precision.

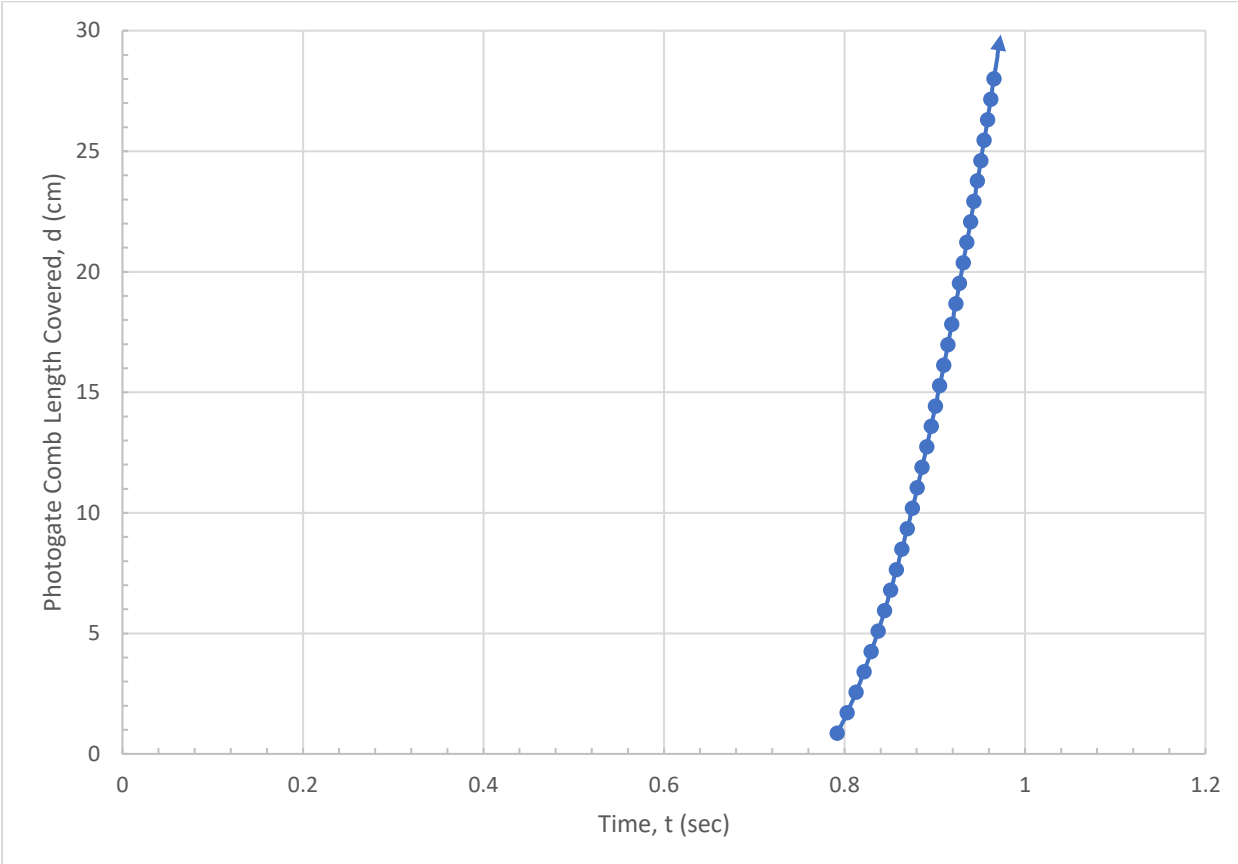

Figure 1.3: Photogate length markers covered over time during free-fall. The graph plots the length of the photogate covered as CAPSTONE records time markers, which forms a second order polynomial best-fit line. By multiplying the second order constant by two, acceleration due to gravity is calculated. Regression analysis of the best-fit line reveals the uncertainty of the calculated value of gravity. The resulting value of is $g = (9.805 \pm 0.0019)\ m/s^2$.

Figure 1.3 shows the results of the photogate comb experiment where a photogate's comb slits pass through a photogate while CAPSTONE records time markers that showed the progression of time and distance as the comb slots accelerated towards the earth due to gravity. CAPSTONE collected the data in the form of block counts and time markers. The block counts are converted to distance by multiplying the amount of block counts at a time marker by the distance per time marker (Δ), which is equivalent to the entire length of the photogate comb divided by the number of comb slits. The first and data point are then removed for the sake of precision. The data produces a second-order-polynomial-best-fit line that indicates the acceleration of the photogate comb due to gravity. By doubling the second order constant because $g = 2a$, g_{best} is calculated while δg is calculated by quadratic regression analysis.